

Exploring the Impact of Haptic Feedback Locations and Mid-air Haptic Technology on Driver's Takeover Performance in Automated Vehicles

Ruiheng Lan
University of Nottingham Ningbo
China
Ningbo, China
suxrl1@nottingham.edu.cn

Xu Sun*
University of Nottingham Ningbo
China
Ningbo, China
xu.sun@nottingham.edu.cn

Linzhen Zhu
University of Michigan
Ann Arbor, Michigan, USA
lzzhu@umich.edu

Qingfeng Wang
University of Nottingham Ningbo
China
Ningbo, China
qingfeng.wang@nottingham.edu.cn

Bingjian Liu
University of Nottingham Ningbo
China
Ningbo, China
bingjian.liu@nottingham.edu.cn

Abstract

Automated vehicles enable drivers to engage in non-driving-related tasks, shifting their roles and introducing challenges for effective human-machine interfaces in managing takeover requests. This study examines the impact of haptic feedback locations (i.e., hand and face) and technologies (i.e., vibrotactile and ultrasonic mid-air haptic) on the driver's takeover performance. In the experiment with 32 participants, results showed that feedback on the face led to significantly faster simple reaction times than on the hand. Additionally, vibrotactile feedback resulted in quicker takeover reaction time and lane change time compared to mid-air haptic technology. These findings highlight the importance of feedback location and technology in safety-critical scenarios and offer insights for designing more intuitive and effective takeover request mechanisms to improve driving safety in future automated vehicles.

CCS Concepts

• **Human-centered computing** → Interaction design; Empirical studies in interaction design..

Keywords

Automated vehicles, Haptic feedback, User experience, Mid-air haptic on the face

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*Corresponding Author: xu.sun@nottingham.edu.cn.

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1 Introduction

Automated vehicles (AVs) promise to transform mobility by allowing drivers to engage in non-driving-related tasks (NDRTs), such as reading or using mobile devices [17, 67]. SAE International classifies Automated Driving Systems into six levels, ranging from Levels 0–2, where drivers must continuously monitor the driving environment, to Levels 4–5, where the driving process is primarily automated [61]. However, until Level 5 is achieved, intermediate Level 3 AVs still require drivers to take control during takeover requests (TORs), a safety-critical transition demanding rapid regaining of situational awareness [42, 46, 49, 78]. Developing effective human-machine interface (HMI) systems to support drivers during these TORs is crucial for ensuring safe operation [3, 31].

Traditional HMI systems primarily rely on visual and auditory feedback to alert drivers of the need for TOR. However, these modalities are prone to limitations under sensory overload, particularly when drivers are engaged in NDRTs. According to Wickens' Multiple Resource Theory [70, 71], overlapping sensory demands degrade task performance, with phenomena like inattention blindness [44] and deafness [43] underscoring the need for alternative feedback mechanisms. Haptic feedback, which directly stimulates the sense of touch, offers a promising solution, as it can be perceived even when drivers are visually or audibly distracted [4, 74]. Its effectiveness has been demonstrated to enhance safety in other fields, such as aviation [41], and to improve response time to hazards and warnings in automotive settings [30, 31, 58, 73].

Despite its advantages, the effectiveness of haptic feedback in AVs depends on its delivery location [20, 33]. Traditional implementations of haptic feedback typically use in-vehicle surfaces in contact with the driver, such as the seat [19], seatbelt [7], steering wheel [2], or pedals [11]. While convenient, these locations have inherent limitations. For example, thick clothing or gloves can attenuate vibrations delivered through the seat or steering wheel, diminishing their effectiveness [48, 64]. More importantly, drivers in AVs often adopt non-driving postures during NDRTs [72], leading

to inconsistent body contact and further reducing feedback reliability [18, 60]. These challenges underscore the need for adaptable, location-independent haptic feedback systems.

To address these limitations, researchers have begun exploring alternative haptic feedback locations and technologies. Wearable devices, such as smartwatches, rings, and smart glasses, can deliver feedback to areas of high tactile sensitivity, like the hands or face [9, 79]. The growing ubiquity of wearables in daily life suggests they could integrate well into AV systems if seamlessly incorporated into the driving experience [40, 77, 79]. However, wearables may impose burdens, as drivers might be unwilling or unable to wear additional devices while driving.

A promising alternative is contactless haptic feedback, particularly ultrasonic mid-air haptic (UMH) technology [35, 38]. UMH delivers tactile sensations without physical contact, making it adaptable to different driver postures and eliminating the need for wearables. This non-invasive approach offers potential for TOR scenarios, as it bypasses user resistance to wearing devices. However, questions remain about its effectiveness in delivering safety-critical signals and drivers' perceptions of this novel technology.

This study investigates the comparative effectiveness of haptic feedback delivered through wearables on the hand and face and UMH technology. By evaluating their impact on driver takeover performance and user experience in a simulated AV environment, the study aims to contribute insights into designing safer, more effective, and user-friendly HMIs for AVs.

The contributions of this paper are:

- **Empirical Comparison:** An empirical comparison of haptic feedback delivered through wearables and UMH feedback on the hand and face within the context of AV HMI design.
- **Exploration of Haptic Feedback Locations and Technologies:** An investigation of the effects of novel feedback locations (hand and face) and technologies (vibrotactile and UMH) on driver takeover performance.
- **Design Recommendations:** Practical recommendations for integrating haptic feedback into AV HMIs to enhance safety.

2 Related Works

2.1 Novel Haptic Feedback Locations in AVs

The hand and face are among the most commonly selected locations for delivering haptic feedback via wearable devices (e.g., smartwatch, wristband, and smart eyewear) [57, 75]. This preference stems from their high tactile sensitivity due to a greater density of tactile receptors compared to other body areas [9, 13]. This heightened sensitivity provides a physiological advantage over traditional vehicle-based feedback locations, such as the seat or steering wheel, in delivering timely and effective TOR signals in AVs [13, 34]. Moreover, the principle of stimulus-response (S-R) compatibility suggests faster response times when the stimulus location aligns with the required action [28]. For TORs, where drivers must promptly regain control, haptic feedback delivered to the hand or face—both directly involved in takeover behaviors such as steering or activating controls—may enhance response speed and effectiveness [5, 77]. For instance, feedback delivered to the hand intuitively aligns with the action of steering.

Recent research has examined haptic feedback on the hand and face during driving. Zheng et al. [77] found that vibrotactile feedback to the face resulted in faster reaction times compared to the hand and shin during a simulated driving task. However, another study reported quicker responses to feedback on the hand than the face in a collision warning system [79]. These findings suggest distinct advantages for both locations but underscore the limited empirical research and lack of consistent conclusions. Notably, no prior studies have directly compared the effectiveness of hand versus face locations for TOR-related haptic feedback.

Addressing these gaps, this study aims to evaluate and compare the efficacy of the hand and face as haptic feedback locations for TORs in AVs. By examining their impact on response times, this research seeks to provide insights into the design of haptic HMIs for AVs.

2.2 Ultrasonic Mid-air Feedback Technology in AVs

UMH technology uses ultrasound transducers to create tactile sensations via focal points generated through phase synchronization [32]. These focal points produce localized pressure on the skin [65], offering contactless interaction with high spatial resolution [24]. UMH has been applied in various fields, including mixed reality [36], digital advertising [23], and HMI design in manual driving vehicles [39, 59], often for representing virtual objects or providing mid-air gesture feedback.

Georgiou et al. [22] introduced a UMH prototype for in-vehicle infotainment, demonstrating its effectiveness in manual driving by significantly reducing the time drivers looked away from the road [39, 59]. Despite its benefits, the application of UMH in AVs, especially as a passive alert system, remains underexplored. Few studies have assessed its effectiveness in AV environments. For example, Hein et al. [26] found UMH to be more pleasant and less intrusive than air-vortex-based feedback and more recognizable than vibrotactile feedback in environments with ambient vibrations [63].

Initially, UMH was primarily applied to the palm [55]. Recent studies, however, have explored its use on the face in contexts such as pedestrian warnings [51], workplace notifications [35], and in-vehicle HMIs [38]. Applying UMH to the face allows drivers to keep their hands free for NDRTs, enhancing the overall AV experience. Despite these advancements, research on UMH applied to the face in AVs and its comparison with traditional contact-based vibrotactile feedback remains limited. This study addresses this gap by comparing UMH and vibrotactile feedback at the hand and face.

2.3 Research Questions

In response to the above-mentioned gaps and challenges in haptic feedback design within AV HMIs, this study addresses two main research questions (RQ):

RQ1: How do the locations of haptic feedback (on the hand and face) influence drivers' takeover performance?

RQ2: What impact do haptic feedback technologies, particularly UMH and vibrotactile, have on takeover performance within AV?

3 Methods

3.1 Participants

Thirty-two participants (16 females; mean age: 23.72 ± 3.47 years) were recruited for this study through online social media platforms and flyers posted around the campus. All participants held valid driving licenses and had at least one year of driving experience (mean = 3.44 ± 2.37 years). Thirteen participants had prior experience with advanced driver assistance systems, such as adaptive cruise control and automatic parking systems. Ethical approval was obtained from the University ethics committee prior to the commencement of the experiment.

3.2 Apparatus and Haptic Feedback Design

The automated driving scenario was developed based on CARLA with a low-fidelity fixed-base driving simulator. Participants controlled the vehicle post-TOR using a Logitech G29 steering control and foot pedal setup. The steering wheel was modified to integrate a UMH transducer array, simulating an AV steering wheel equipped with UMH feedback.

Vibrotactile feedback was delivered through a vibration motor module (24mm × 25mm), controlled wirelessly via an Arduino board (Seeed Studio XIAO ESP32S3). Operating at 200Hz with a 5V power supply, the motor targeted the optimal sensitivity range of human tactile perception [29]. UMH feedback was generated by the STRATOS Explore device, featuring a 16×16 array of ultrasonic transducers. This setup provided UMH feedback within a 60-degree field at distances up to 800mm. Lateral modulation (LM) was employed to enhance sensation, with LM frequency set at 70Hz and amplitude at 12mm [38, 66]. An Intel RealSense D435 depth camera, mounted above the transducer array, tracked participants' faces at 30Hz [38]. The array and camera were centrally mounted on the steering wheel to direct UMH feedback to the face, with no fixed head position required. For hand-focused UMH feedback, the device was housed in a 3D-printed hand support with a 6cm × 6cm aperture to deliver stimuli at an optimal distance of 20cm.

In this study, four feedback designs tailored for future AV systems were developed, incorporating two locations and two haptic technologies (Figure 1):

- 1) Vibrotactile feedback on the hand (VH): A vibrotactile module was secured to the participant's right wrist with medical tape, simulating a wearable wristband.
- 2) UMH feedback on the hand (UH): Applied to the center of the palm, exploiting its high tactile sensitivity and large area conducive to UMH reception. This location aligns with the mid-air gesture interface concept commonly explored in AV HMIs, enhancing its relevance for future applications [12, 27].
- 3) Vibrotactile feedback on the face (VF): Considering the likelihood of integrating vibrotactile feedback into devices such as smart glasses or wireless earphones, the module was attached to the upper jaw [77].
- 4) UMH feedback on the face (UF): Given the power limitations of current UMH devices, feedback was directed to the center of the lip, a region noted for its heightened sensitivity and

perceived intensity of UMH feedback. This location has been identified as optimal for delivering alert signals [35, 38].

These feedback designs were grounded in S-R compatibility principles, ensuring alignment with the specific actions required during the takeover task [28]. Haptic feedback on the hand was selected for its compatibility with steering wheel maneuvers, while feedback on the face was chosen to correspond with the natural upward head movement involved in disengaging from NDRTs. To further enhance spatial compatibility [68] and minimize reaction times, all feedback (except UF, due to technical limitations) was delivered on the right side of the body, consistent with the visual instruction to change to the right lane. Additionally, only right-handed participants were included in the study to maintain consistency and avoid variabilities associated with hand dominance.

3.3 Experiment Design

This study employed a 2×2 factorial within-subject design to analyze the effects of two feedback locations (hand and face) and two haptic technologies (vibrotactile as contact-based and UMH as contactless-based). Participants experienced four TOR feedback designs across four automated driving scenarios. To mitigate order effects and ensure robust results, the feedback designs were counterbalanced using a Latin square design [76].

3.3.1 Takeover Scenarios. The driving simulation was conducted on an eight-lane road with four lanes in each direction (Figure 2). The ego-vehicle began in the far-right emergency lane and automatically transitioned to the third lane, maintaining a constant speed of 60 km/h. To isolate the effects of the feedback designs, no other vehicles were present during the simulation. At a random point between 2 and 3 minutes after the start of the scenario, while the ego-vehicle was traveling along a straight road, the TOR was initiated. Upon receiving the TOR, participants were required to disengage from the NDRT and resume control of the vehicle by pressing either the accelerator or brake pedal. After taking over, participants were instructed to execute a lane change to the right-adjacent lane following the visual indicator [15]. The time taken to complete the lane change after receiving the TOR was recorded as a measure of the driver's information processing speed [19]. Once the lane change was completed, participants continued driving manually at a set speed for 30 seconds before the scenario concluded.

3.3.2 1-back Task. The NDRT selected for this study was the 1-back task [37], a widely used cognitive task in automotive research. This task demands cognitive resources comparable to those required for operating in-vehicle infotainment systems [6, 14, 69], making it an appropriate choice for simulating driver engagement in NDRTs during automated driving [47, 54]. As shown in Figure 1, numbers between 0 and 9 were presented visually on a smartphone display attached to the bottom of the steering wheel. Each number appeared for 500 ms, with a 3-second interval between successive numbers. Participants were instructed to press a button on the armrest with their left hand as quickly and accurately as possible whenever the current number matched the previous (1-back) number. This setup ensured that participants' right hands remained in a fixed location during UMH conditions.

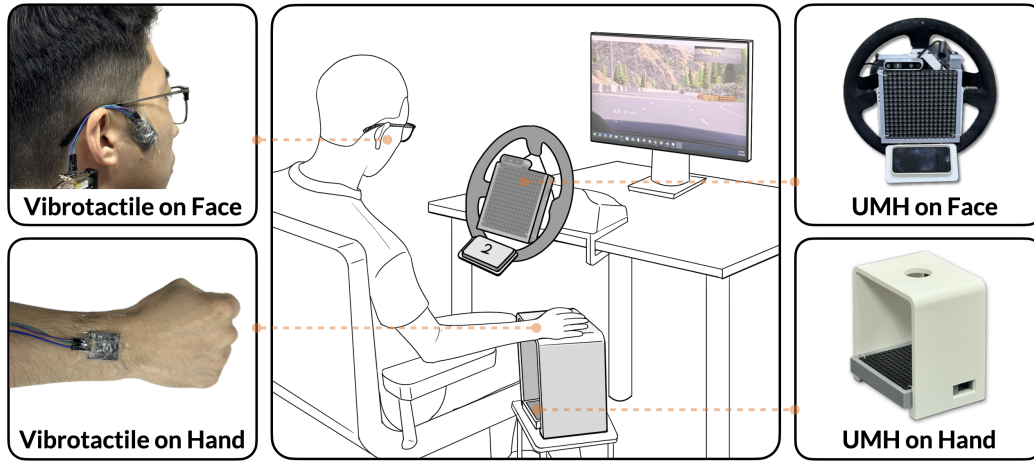


Figure 1: Four haptic feedback designs used in the experiment.

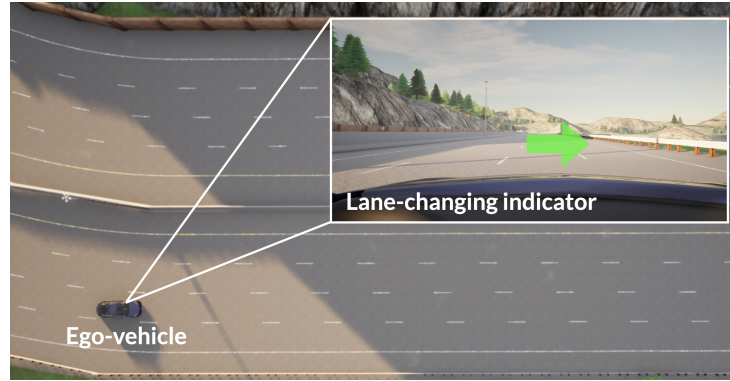


Figure 2: Takeover scenario.

3.4 Procedure

Upon arrival, participants were briefed on the study and provided informed consent. They then completed a demographics questionnaire and were introduced to the four haptic feedback designs. Before formal testing, participants spent 3-5 minutes familiarizing themselves with the driving simulator and manually operating the automated vehicle in the simulated environment.

The formal test consisted of four counterbalanced blocks, each evaluating a specific haptic feedback design. In each block, participants first performed a reaction time task, pressing a button with their left hand upon detecting the haptic stimulus. This task was repeated five times, with the average serving as the simple reaction time (SRT). Participants then engaged in a simulated automated driving scenario (as described in 3.3.1) while performing a 1-back task, which was conducted once per block. To reduce auditory distractions, especially during UMH feedback blocks, participants wore 3M noise-isolating earplugs. Each block lasted approximately 10 minutes, with the entire study taking about 60 minutes. Participants received a \$10 gift as appreciation for their time.

3.5 Measurement and Data Analysis

Three objective metrics were used to evaluate drivers' takeover performance: SRT, takeover reaction time (TRT), and lane change time (LCT). TRT was measured from the onset of the TOR signal to the initial contact with the brake pedal [16], while LCT was calculated as the interval between the TOR presentation and the successful lane change completion [19]. These objective data were analyzed using a two-way repeated measures ANOVA in SPSS 28. Post-hoc pairwise comparisons with Bonferroni corrections were conducted.

4 Results

As shown in Figure 3, the impact of haptic feedback technology on reaction times was significant across several measures. Specifically, vibrotactile feedback significantly reduced SRT compared to UMH feedback ($F(1, 31) = 456.202, p < 0.001, \eta^2 = 0.936$). This effect also extended to TRT ($F(1, 31) = 20.582, p < 0.001, \eta^2 = 0.399$), and LCT ($F(1, 31) = 10.336, p = 0.003, \eta^2 = 0.250$), demonstrating the efficacy of vibrotactile feedback in enhancing rapid response capabilities in AV contexts. Furthermore, a noteworthy interaction effect between

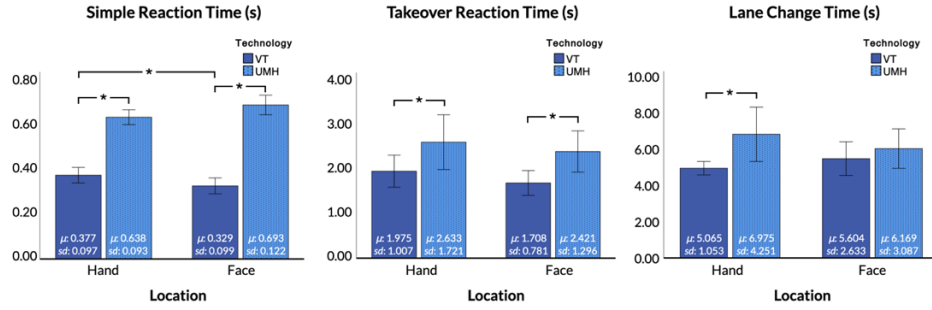


Figure 3: Summary of the result on reaction time and lane change time across four haptic feedback designs. Bars represent the mean; error bars represent the 95% confidence intervals. VT = Vibrotactile; UMH = Ultrasonic Mid-air Haptic; μ = mean value; sd = standard deviation; * indicates a significant difference between groups ($p < 0.05$).

the feedback location and technology on SRT ($F(1, 31) = 13.730$, $p < 0.001$, $\eta^2 = 0.307$) revealed that reactions were significantly faster when vibrotactile feedback was applied to the face compared to the hand ($p < 0.001$), whereas UMH feedback was more effective on the hand than the face ($p = 0.021$).

5 Discussion

The study's results demonstrated that haptic feedback technologies significantly influenced drivers' TRT and LCT. Contact-based feedback consistently resulted in significantly quicker TRT and LCT across both tested feedback locations compared to UMH feedback, and this trend extended to SRT, indicating a fundamental difference in reaction efficiency between the two technologies. Several factors may explain this disparity. First, the inherent delay in UMH feedback, caused by the time required for ultrasound waves to reach the skin, likely contributes to slower reaction times. Additionally, the weaker tactile sensation produced by UMH may further delay responses. This aligns with findings by Conway et al. [8], who observed that reduced vibration intensity negatively impacts performance. Unlike vibrotactile feedback, which directly stimulates subcutaneous mechanoreceptors (e.g., Pacinian corpuscles responsible for vibration detection [10]) through physical contact, UMH generates tactile sensations by generating air pressure waves, stimulating mechanoreceptors at a less effective level [65]. These differences in stimulation mechanisms likely result in less effective haptic perception for UMH compared to vibrotactile feedback.

The varying tactile sensitivities of the stimulation locations may also influence the observed differences in reaction times between vibrotactile and UMH feedback. For the hand condition, UMH feedback was delivered to the palm, a region with higher receptor density and larger receptive fields compared to the wrist, where vibrotactile feedback was applied [45]. Similarly, for the face condition, UMH feedback was directed to the lip, an area of heightened sensitivity [62], whereas vibrotactile feedback was applied to the upper jaw. Since tactile stimuli are more easily perceived in areas with higher sensitivity, the actual difference in reaction times between the two technologies may be larger than observed in this study. Future research could address this confound by applying both feedback technologies to identical locations.

The difference between technologies was particularly pronounced in the TOR scenario, where urgency is critical. The mean difference in TRTs (615 ms) was substantially larger than in SRTs (312 ms), highlighting the superior efficacy of contact-based feedback in situations requiring swift control transitions. Furthermore, a significant interaction between feedback location and technology was observed in SRT. Contact-based feedback was faster on the face than on the hand, consistent with findings by Zheng et al. [77], who reported quicker SRT for facial feedback compared to wrist feedback. This difference may result from the shorter neural pathway between the face and the brain, facilitating faster processing [25]. Another explanation could be the auditory cues generated by vibrotactile feedback via bone conduction [1], which some participants reported as noticeable and occasionally uncomfortable during the experiment. Conversely, the response from UMH feedback did not align with the neural proximity hypothesis. Instead, the larger stimulated area on the hand may have enhanced the perceived intensity of UMH feedback through spatial summation at the receptor level, potentially accelerating response times [52].

6 Design Implication

This study demonstrates that different haptic feedback designs significantly impact drivers' takeover performance. Based on these findings, we highlight several design paradigms and research opportunities to enhance AV HMIs.

6.1 Haptic Feedback Location Selection

The placement of haptic feedback should be context-dependent to ensure a balance between user alertness and comfort. Both the hands and face were effective for delivering TOR signals, showing similar response times. Designers are encouraged to explore either location for developing specific forms of TOR devices. In time-critical scenarios, vibrotactile feedback on the face is recommended, as it produces the fastest response times.

6.2 Haptic Feedback Technology Selection

For TOR scenarios, contact-based haptic feedback is recommended for superior performance. Conversely, UMH feedback, noted for its non-contact convenience and slower response times, is better suited for less critical applications [38]. For example, it can be

effectively used in in-vehicle infotainment systems to confirm user interactions, such as volume adjustments [38, 39].

7 Limitations and Future Work

This exploratory study, focused only on one basic TOR scenario with participants engaged in a standardized 1-back NDRT, highlights the nuances of haptic feedback in AV HMIs. Given that different NDRTs can significantly affect takeover performance [50] and the diversity of NDRTs in real AV contexts [67], future research should delve deeper into how various NDRTs influence the efficacy of different haptic feedback designs. To streamline the experimental design and minimize potential confounding factors, this study focused exclusively on haptic feedback for TOR. However, existing literature suggests that multimodal feedback may facilitate faster responses than haptic feedback alone [21, 31, 53]. Therefore, investigating the integration of haptic feedback with other sensory modalities, and examining their interactions with different feedback locations and technologies, represents a promising direction for future research. This approach could potentially enhance the responsiveness and intuitiveness of HMIs in safety-critical situations, offering a more comprehensive understanding of multimodal feedback in AV environments. Additionally, while this study did not systematically analyze participant comments, user experience insights are critical as they directly impact AV safety and adoption [56]. Future research should adopt a more structured approach to capturing and analyzing user feedback, which could provide valuable insights into how different haptic feedback designs influence drivers' emotional responses and their attitudes toward AVs.

8 Conclusion

This study investigated the impact of haptic feedback locations and technologies on drivers' takeover performance in AV HMIs. The results highlighted that contact-based vibrotactile feedback, particularly on the face, significantly improved response times. Face-based feedback also led to faster SRT compared to hand-based feedback. These findings underscore the importance of feedback location and technology in designing effective AV HMIs. Future research should explore these parameters further to create more intuitive, safety-enhancing, and user-friendly HMIs for AVs.

References

- [1] Tian Bao, Lydia Su, Catherine Kinnaird, Mohammed Kabeto, Peter B. Shull, and Kathleen H. Sienko. 2019. Vibrotactile display design: Quantifying the importance of age and various factors on reaction times. *PLOS ONE* 14, 8: e0219737. <https://doi.org/10.1371/journal.pone.0219737>
- [2] Shadan Sadeghian Borojeni, Torben Wallbaum, Wilko Heuten, and Susanne Boll. 2017. Comparing Shape-Changing and Vibro-Tactile Steering Wheels for Take-Over Requests in Highly Automated Driving. In *Proceedings of the 9th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (AutomotiveUI '17), 221–225. <https://doi.org/10.1145/3122986.3123003>
- [3] Oliver Carsten and Marieke H. Martens. 2019. How can humans understand their automated cars? HMI principles, problems and solutions. *Cognition, Technology & Work* 21, 1: 3–20. <https://doi.org/10.1007/s10111-018-0484-0>
- [4] Chunlei Chai, Yu Lei, Haoran Wei, Changxu Wu, Wei Zhang, Preben Hansen, Hao Fan, and Jinlei Shi. 2024. The effects of various auditory takeover requests: A simulated driving study considering the modality of non-driving-related tasks. *Applied Ergonomics* 118: 104252. <https://doi.org/10.1016/j.apergo.2024.104252>
- [5] Ken WL Chan and Alan HS Chan. 2011. Spatial stimulus response compatibility for a horizontal visual display with hand and foot controls. *Ergonomics* 54, 3: 233–245.
- [6] Damee Choi, Toshihisa Sato, Takafumi Ando, Takashi Abe, Motoyuki Akamatsu, and Satoshi Kitazaki. 2020. Effects of cognitive and visual loads on driving performance after take-over request (TOR) in automated driving. *Applied Ergonomics* 85: 103074. <https://doi.org/10.1016/j.apergo.2020.103074>
- [7] Jaemin Chun, In Lee, Gunhyuk Park, Jongman Seo, Seungmoon Choi, and Sung H. Han. 2013. Efficacy of haptic blind spot warnings applied through a steering wheel or a seatbelt. *Transportation Research Part F: Traffic Psychology and Behaviour* 21: 231–241. <https://doi.org/10.1016/j.trf.2013.09.014>
- [8] GE Conway, JL Szalma, and PA Hancock. 2007. A quantitative meta-analytic examination of whole-body vibration effects on human performance. *Ergonomics* 50, 2: 228–245.
- [9] Giulia Corniani and Hannes P. Saal. 2020. Tactile innervation densities across the whole body. *Journal of Neurophysiology* 124, 4: 1229–1240. <https://doi.org/10.1152/jn.00313.2020>
- [10] J Dargahi and S Najarian. 2004. Human tactile perception as a standard for artificial tactile sensing—a review. *The International Journal of Medical Robotics and Computer Assisted Surgery* 1, 1: 23–35. <https://doi.org/10.1002/rcs.3>
- [11] Helios De Rosario, Marcos Louredo, Iñaki Díaz, Andrés Soler, Jorge Juan Gil, José S. Solaz, and Jordi Jornet. 2010. Efficacy and feeling of a vibrotactile Frontal Collision Warning implemented in a haptic pedal. *Transportation Research Part F: Traffic Psychology and Behaviour* 13, 2: 80–91. <https://doi.org/10.1016/j.trf.2009.11.003>
- [12] Henrik Detjen, Stefan Geisler, and Stefan Schneegass. 2021. Driving as Side Task: Exploring Intuitive Input Modalities for Multitasking in Automated Vehicles. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*, 1–6. <https://doi.org/10.1145/3411763.3451803>
- [13] Nem Khan Dim and Xiangshi Ren. 2017. Investigation of suitable body parts for wearable vibration feedback in walking navigation. *International Journal of Human-Computer Studies* 97: 34–44. <https://doi.org/10.1016/j.ijhcs.2016.08.002>
- [14] Yaohan Ding, Lesong Jia, and Na Du. 2024. One Size Does Not Fit All: Designing and Evaluating Criticality-Adaptive Displays in Highly Automated Vehicles. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 1–15. <https://doi.org/10.1145/3613904.3642648>
- [15] Na Du, Feng Zhou, Dawn Tilbury, Lionel Peter Robert, and X. Jessie Yang. 2021. Designing Alert Systems in Takeover Transitions: The Effects of Display Information and Modality. In *13th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*, 173–180. <https://doi.org/10.1145/3409118.3475155>
- [16] Society of Automotive Engineers. 2015. Operational definitions of driving performance measures and statistics. *Surface Vehicle Recommended Practice J2944*.
- [17] Daniel J Fagnant and Kara Kockelman. 2015. Preparing a nation for autonomous vehicles: opportunities, barriers and policy recommendations. *Transportation Research Part A: Policy and Practice* 77: 167–181.
- [18] Martin Fleischer and Si Chen. 2020. How Do We Sit When Our Car Drives for Us? In *Digital Human Modeling and Applications in Health, Safety, Ergonomics and Risk Management. Posture, Motion and Health*, Vincent G. Duffy (ed.). Springer International Publishing, Cham, 33–49. https://doi.org/10.1007/978-3-030-49904-4_3
- [19] G. Huang and B. J. Pitts. 2022. To Inform or to Instruct? An Evaluation of Meaningful Vibrotactile Patterns to Support Automated Vehicle Takeover Performance. *IEEE Transactions on Human-Machine Systems*: 1–10. <https://doi.org/10.1109/THMS.2022.3205880>
- [20] Yoren Gaffary and Anatole Lécuyer. 2018. The Use of Haptic and Tactile Information in the Car to Improve Driving Safety: A Review of Current Technologies. *Frontiers in ICT* 5: 5. <https://doi.org/10.3389/fict.2018.00005>
- [21] C Geitner, F Biondi, L Skrypchuk, P Jennings, and S Birrell. 2019. The comparison of auditory, tactile, and multimodal warnings for the effective communication of unexpected events during an automated driving scenario. *TRANSPORTATION RESEARCH PART F-TRAFFIC PSYCHOLOGY AND BEHAVIOUR* 65: 23–33. <https://doi.org/10.1016/j.trf.2019.06.011>
- [22] Orestis Georgiou, Valerio Biscione, Adam Harwood, Daniel Griffiths, Marcello Giordano, Ben Long, and Tom Carter. 2017. Haptic In-Vehicle Gesture Controls. In *Proceedings of the 9th International Conference on Automotive User Interfaces and Interactive Vehicular Applications Adjunct*, 233–238. <https://doi.org/10.1145/3131726.3132045>
- [23] Orestis Georgiou, Hannah Limerick, Loïc Corenthy, Mark Perry, Mykola Maksymenko, Sam Frish, Jörg Müller, Myroslav Bachynskyi, and Jin Ryong Kim. 2019. Mid-air haptic interfaces for interactive digital signage and kiosks. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–9.
- [24] Hyunjae Gil, Hyungki Son, Jin Ryong Kim, and Ian Oakley. 2018. Whiskers: Exploring the Use of Ultrasonic Haptic Cues on the Face. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 1–13. <https://doi.org/10.1145/3173574.3174232>
- [25] Vanessa Harrar and Laurence R Harris. 2005. Simultaneity constancy: detecting events with touch and vision. *Experimental Brain Research* 166: 465–473.
- [26] Philipp Hein, Dursune Gönültaş, Max Bernhagen, André Dettmann, and Angelika C. Bullinger. 2022. Mid-Air Haptic Cues for Safety-Critical Communication: (Re-)Exploring Air-Vortex Rings in Human-Machine Interaction. In *Mensch und Computer 2022*, 41–51. <https://doi.org/10.1145/3543758.3543949>

- [27] Jahani F. Hessam, Massimo Zancanaro, Manolya Kavakli, and Mark Billinghurst. 2017. Towards optimization of mid-air gestures for in-vehicle interactions. In *Proceedings of the 29th Australian Conference on Computer-Human Interaction*, 126–134. <https://doi.org/10.1145/3152771.3152785>
- [28] Cristy Ho and Charles Spence. 2014. Effectively responding to tactile stimulation: Do homologous cue and effector locations really matter? *Acta Psychologica* 151: 32–39. <https://doi.org/10.1016/j.actpsy.2014.05.014>
- [29] Cristy Ho, Hong Z Tan, and Charles Spence. 2006. The differential effect of vibrotactile and auditory cues on visual spatial attention. *Ergonomics* 49, 7: 724–738.
- [30] GJ Huang and BJ Pitts. 2022. The effects of age and physical exercise on multimodal signal responses: Implications for semi-autonomous vehicle takeover requests. *APPLIED ERGONOMICS* 98. <https://doi.org/10.1016/j.apergo.2021.103595>
- [31] GJ Huang and BJ Pitts. 2022. Takeover requests for automated driving: The effects of signal direction, lead time, and modality on takeover performance. *ACCIDENT ANALYSIS AND PREVENTION* 165. <https://doi.org/10.1016/j.aap.2021.106534>
- [32] Takayuki Iwamoto, Mari Tatezono, and Hiroyuki Shinoda. 2008. Non-contact method for producing tactile sensation using airborne ultrasound. In *Haptics: Perception, Devices and Scenarios: 6th International Conference, EuroHaptics 2008 Madrid, Spain, June 10-13, 2008 Proceedings* 6, 504–513.
- [33] Pascal Jansen, Mark Colley, and Enrico Rukzio. 2022. A Design Space for Human Sensor and Actuator Focused In-Vehicle Interaction Based on a Systematic Literature Review. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 6, 2: 1–51. <https://doi.org/10.1145/3534617>
- [34] Reinier J. Jansen, Angelica M. Tinga, Rins De Zwart, and Sander T. Van Der Kint. 2022. Devil in the details: Systematic review of TOR signals in automated driving with a generic classification framework. *Transportation Research Part F: Traffic Psychology and Behaviour* 91: 274–328. <https://doi.org/10.1016/j.trf.2022.10.009>
- [35] Arata Jingu, Takaaki Kamigaki, Masahiro Fujiwara, Yasutoshi Makino, and Hiroyuki Shinoda. 2021. LipNotif: Use of Lips as a Non-Contact Tactile Notification Interface Based on Ultrasonic Tactile Presentation. In *The 34th Annual ACM Symposium on User Interface Software and Technology*, 13–23. <https://doi.org/10.1145/3472749.3474732>
- [36] Cédric Kervégant, Félix Raymond, Delphine Graeff, and Julien Castet. 2017. Touch hologram in mid-air. In *ACM SIGGRAPH 2017 emerging technologies*. 1–2.
- [37] Wayne K Kirchner. 1958. Age differences in short-term retention of rapidly changing information. *Journal of experimental psychology* 55, 4: 352.
- [38] Ruiheng Lan, Xu Sun, Qingfeng Wang, and Bingjian Liu. 2024. Ultrasonic Mid-Air Haptics on the Face: Effects of Lateral Modulation Frequency and Amplitude on Users' Responses. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*.
- [39] David R. Large, Kyle Harrington, Gary Burnett, and Orestis Georgiou. 2019. Feel the noise: Mid-air ultrasound haptics as a novel human-vehicle interaction paradigm. *Applied Ergonomics* 81: 102909. <https://doi.org/10.1016/j.apergo.2019.102909>
- [40] Wei-Hsiang Lo and Gaojian Huang. 2024. A Survey on Perceptions of Smart-watch Haptic Feedback for Enhancing Automated Vehicle Takeover Decisions. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*: 10711813241275929. <https://doi.org/10.1177/10711813241275929>
- [41] Luis Lutnyk, David Rudi, Emanuel Meier, Peter Kiefer, and Martin Raubal. 2022. FlyBrate: Evaluating Vibrotactile Cues for Simulated Flight. *International Journal of Human-Computer Interaction*: 1–18. <https://doi.org/10.1080/10447318.2022.2075627>
- [42] Shu Ma, Wei Zhang, Zhen Yang, Chunyan Kang, Changxu Wu, Chunlei Chai, Jinlei Shi, Yilin Zeng, and Hongting Li. 2021. Take over Gradually in Conditional Automated Driving: The Effect of Two-stage Warning Systems on Situation Awareness, Driving Stress, Takeover Performance, and Acceptance. *International Journal of Human-Computer Interaction* 37, 4: 352–362. <https://doi.org/10.1080/10447318.2020.1860514>
- [43] James SP Macdonald and Nilli Lavie. 2011. Visual perceptual load induces inattentional deafness. *Attention, Perception, & Psychophysics* 73: 1780–1789.
- [44] Arien Mack. 2003. Inattention blindness: Looking without seeing. *Current directions in psychological science* 12, 5: 180–184.
- [45] Flavia Mancini, Armando Bauleo, Jonathan Cole, Fausta Lui, Carlo A. Porro, Patrick Haggard, and Gian Domenico Iannetti. 2014. Whole-body mapping of spatial acuity for pain and touch. *Annals of Neurology* 75, 6: 917–924. <https://doi.org/10.1002/ana.24179>
- [46] Anthony D McDonald, Hananeh Alambeigi, Johan Engström, Gustav Markkula, Tobias Vogelwohl, Jarrett Dunne, and Norbert Yuma. 2019. Toward Computational Simulations of Behavior During Automated Driving Takeovers: A Review of the Empirical and Modeling Literatures. *Human Factors*.
- [47] Bruce Mehler, Bryan Reimer, and Jeffery A Dusek. 2011. MIT AgeLab delayed digit recall task (n-back). Cambridge, MA: Massachusetts Institute of Technology 17.
- [48] Fanxing Meng and Charles Spence. 2015. Tactile warning signals for in-vehicle systems. *Accident Analysis & Prevention* 75: 333–346. <https://doi.org/10.1016/j.aap.2014.12.013>
- [49] Joel A Miller, Soodeh Nikan, and Mohamed H Zaki. 2024. Navigating the Handover: Reviewing Takeover Requests in Level 3 Autonomous Vehicles. *IEEE Open Journal of Vehicular Technology*.
- [50] Saad Minhas, Aura Hernandez-Sabate, Shoaib Ehsan, and Klaus D. McDonald-Maier. 2022. Effects of Non-Driving Related Tasks During Self-Driving Mode. *IEEE Transactions on Intelligent Transportation Systems* 23, 2: 1391–1399. <https://doi.org/10.1109/TITS.2020.3025542>
- [51] Saya Mizutani, Masahiro Fujiwara, Yasutoshi Makino, and Hiroyuki Shinoda. 2019. Thresholds of Haptic and Auditory Perception in Midair Facial Stimulation. In *2019 IEEE International Symposium on Haptic, Audio and Visual Environments and Games (HAVE)*, 1–6. <https://doi.org/10.1109/HAVE.2019.8920999>
- [52] Dario Pittera, Orestis Georgiou, Abdenaceur Abdouni, and William Frier. 2022. “I Can Feel It Coming in the Hairs Tonight”: Characterising Mid-Air Haptics on the Hairy Parts of the Skin. *IEEE Transactions on Haptics* 15, 1: 188–199. <https://doi.org/10.1109/TOH.2021.3110722>
- [53] Brandon J. Pitts and Nadine Sarter. 2018. What You Don't Notice Can Harm You: Age-Related Differences in Detecting Concurrent Visual, Auditory, and Tactile Cues. *Human Factors: The Journal of the Human Factors and Ergonomics Society* 60, 4: 445–464. <https://doi.org/10.1177/0018720818759102>
- [54] Jonas Radlmayr, Christian Gold, Lutz Lorenz, Mehdi Farid, and Klaus Bengler. 2014. How Traffic Situations and Non-Driving Related Tasks Affect the Take-Over Quality in Highly Automated Driving. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting* 58, 1: 2063–2067. <https://doi.org/10.1177/1541931214581434>
- [55] Ismo Rakkolainen, Euan Freeman, Antti Sand, Roope Raisamo, and Stephen Brewster. 2021. A Survey of Mid-Air Ultrasound Haptics and Its Applications. *IEEE Transactions on Haptics* 14, 1: 2–19. <https://doi.org/10.1109/TOH.2020.3018754>
- [56] Christina Rödel, Susanne Stadler, Alexander Meschtscherjakov, and Manfred Tscheligi. 2014. Towards Autonomous Cars: The Effect of Autonomy Levels on Acceptance and User Experience. In *Proceedings of the 6th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*, 1–8. <https://doi.org/10.1145/2667317.2667330>
- [57] Suranga Seneviratne, Yining Hu, Tham Nguyen, Guohao Lan, Sara Khalifa, Kanachana Thilakarathna, Mahbub Hassan, and Aruna Seneviratne. 2017. A Survey of Wearable Devices and Challenges. *IEEE Communications Surveys & Tutorials* 19, 4: 2573–2620. <https://doi.org/10.1109/COMST.2017.2731979>
- [58] Farzaneh Shahini and Maryam Zahabi. 2022. Effects of levels of automation and non-driving related tasks on driver performance and workload: A review of literature and meta-analysis. *Applied Ergonomics* 104: 103824. <https://doi.org/10.1016/j.apergo.2022.103824>
- [59] Gözel Shakeri, John H. Williamson, and Stephen Brewster. 2018. May the Force Be with You: Ultrasound Haptic Feedback for Mid-Air Gesture Interaction in Cars. In *Proceedings of the 10th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*, 1–10. <https://doi.org/10.1145/3239060.3239081>
- [60] Shabahang Shayegan, George Beurier, Lisa Denninger, and Xuguang Wang. 2023. Preferred Postures for Doing Non-driving Related Activities in Reclined Seating Configurations in Highly Automated Vehicles. In *Advances in Digital Human Modeling*, Sofia Scataglini, Gregor Harih, Wim Saeys and Steven Truijen (eds.). Springer Nature Switzerland, Cham, 238–247. https://doi.org/10.1007/978-3-031-37848-5_27
- [61] Elisabeth Shi, Tom Michael Gasser, Andre Seeck, and Rico Auerswald. 2020. The Principles of Operation Framework: A Comprehensive Classification Concept for Automated Driving Functions. *SAE International Journal of Connected and Automated Vehicles* 3, 1: 12-03-01–0003. <https://doi.org/10.4271/12-03-01-0003>
- [62] Maria Z Siemionow, Bahar Bassiri Gharb, and Antonio Rampazzo. 2011. The face as a sensory organ. *The Know-How of Face Transplantation*: 11–23.
- [63] Oleg Spakov, Ahmed Farooq, Hanna Venesvirta, Arto Hippula, Veikko Surakka, and Roope Raisamo. 2022. Ultrasound feedback for mid-air gesture interaction in vibrating environment. In *Human Interaction and Emerging Technologies: Artificial Intelligence & Future Applications*.
- [64] C. Spence and C. Ho. 2008. Tactile and Multisensory Spatial Warning Signals for Drivers. *IEEE Transactions on Haptics* 1, 2: 121–129. <https://doi.org/10.1109/TOH.2008.14>
- [65] Chongyang Sun, Weizhi Nai, and Xiaoying Sun. 2019. Tactile sensitivity in ultrasonic haptics: Do different parts of hand and different rendering methods have an impact on perceptual threshold? *Virtual Reality & Intelligent Hardware* 1, 3: 265–275. <https://doi.org/10.3724/SP.J.2096-5796.2019.0009>
- [66] Ryoko Takahashi, Keisuke Hasegawa, and Hiroyuki Shinoda. 2020. Tactile Stimulation by Repetitive Lateral Movement of Midair Ultrasound Focus. *IEEE Transactions on Haptics* 13, 2: 334–342. <https://doi.org/10.1109/TOH.2019.2946136>
- [67] Pinyan Tang, Xu Sun, and Shi Cao. 2020. Investigating user activities and the corresponding requirements for information and functions in autonomous vehicles of the future. *International Journal of Industrial Ergonomics* 80: 103044. <https://doi.org/10.1016/j.ergon.2020.103044>
- [68] Steve N.H. Tsang and Alan H.S. Chan. 2018. Tracking and discrete dual task performance for different visual spatial stimulus-response mappings with focal and ambient vision. *Applied Ergonomics* 67: 39–49. <https://doi.org/10.1016/j.apergo.2017.09.004>
- [69] Nikolai Von Janczewski, Jennifer Wittmann, Arnd Engeln, Martin Baumann, and Lutz Krauß. 2021. A meta-analysis of the n-back task while driving and its effects

- on cognitive workload. *Transportation Research Part F: Traffic Psychology and Behaviour* 76: 269–285. <https://doi.org/10.1016/j.trf.2020.11.014>
- [70] Christopher D. Wickens. 2002. Multiple resources and performance prediction. *Theoretical Issues in Ergonomics Science* 3, 2: 159–177. <https://doi.org/10.1080/14639220210123806>
- [71] Christopher D. Wickens. 2008. Multiple Resources and Mental Workload. *Human Factors: The Journal of the Human Factors and Ergonomics Society* 50, 3: 449–455. <https://doi.org/10.1518/001872008X288394>
- [72] Yucheng Yang, Jan Niklas Klinkner, and Klaus Bengler. 2019. How Will the Driver Sit in an Automated Vehicle? – The Qualitative and Quantitative Descriptions of Non-Driving Postures (NDPs) When Non-Driving-Related-Tasks (NDRTs) Are Conducted. In *Proceedings of the 20th Congress of the International Ergonomics Association (IEA 2018)*, Sebastiano Bagnara, Riccardo Tartaglia, Sara Albolino, Thomas Alexander and Yushi Fujita (eds.). Springer International Publishing, Cham, 409–420. https://doi.org/10.1007/978-3-319-96074-6_44
- [73] SH Yoon, YW Kim, and YG Ji. 2019. The effects of takeover request modalities on highly automated car control transitions. *ACCIDENT ANALYSIS AND PREVENTION* 123: 150–158. <https://doi.org/10.1016/j.aap.2018.11.018>
- [74] Sol Hee Yoon and Yong Gu Ji. 2019. Non-driving-related tasks, workload, and takeover performance in highly automated driving contexts. *Transportation Research Part F: Traffic Psychology and Behaviour* 60: 620–631. <https://doi.org/10.1016/j.trf.2018.11.015>
- [75] Clint Zeagler. 2017. Where to wear it: functional, technical, and social considerations in on-body location for wearable technology 20 years of designing for wearability. In *Proceedings of the 2017 ACM International Symposium on Wearable Computers*, 150–157. <https://doi.org/10.1145/3123021.3123042>
- [76] René Zeelenberg and Diane Pecher. 2015. A method for simultaneously counterbalancing condition order and assignment of stimulus materials to conditions. *Behavior research methods* 47: 127–133.
- [77] Jingyue Zheng, Tingru Zhang, Liang Ma, Yulun Wu, and Wei Zhang. 2021. Vibration warning design for reaction time reduction under the environment of intelligent connected vehicles. *Applied Ergonomics* 96: 103490. <https://doi.org/10.1016/j.apergo.2021.103490>
- [78] Feng Zhou, X. Jessie Yang, and Joost C. F. De Winter. 2022. Using Eye-Tracking Data to Predict Situation Awareness in Real Time During Takeover Transitions in Conditionally Automated Driving. *IEEE Transactions on Intelligent Transportation Systems* 23, 3: 2284–2295. <https://doi.org/10.1109/TITS.2021.3069776>
- [79] Ao Zhu, Shi Cao, Han Yao, Murtuza Jadliwala, and Jibo He. 2020. Can wearable devices facilitate a driver's brake response time in a classic car-following task? *IEEE Access* 8: 40081–40087.